Tiny Worlds: Evaluation 3

Prototype 2b | From anatomy discovery to habitat navigation and aphid identification

RESEARCH FOCUS Does the immersive learning sequence support insect recognition while remaining independently usable through navigation and XR grabbing?

1 Evaluation frame

Prototype 2b connects ladybird body exploration, player-guided travel to a habitat, learning cards and an aphid video, and an XR food-search task. Five participants (P01–P05), all reporting some VR experience, completed moderated sessions. Each performed seven tasks on Meta Quest 2, answered identical two-item knowledge questions before and after the experience, rated seven aspects on a five-point scale, and gave interview comments.

The test plan assessed five intentions: O1 progression clarity; O2 ladybird navigation; O3 aphid recognition; O4 correct XR grabbing; and O5 clear, comfortable feedback. Since the plan stated no fixed numerical cutoff, success is judged descriptively through independence, mistakes, correct answers, ratings and critical incidents rather than invented post-test targets.

2 Factual results

Measure	Recorded outcome	Interpretive boundary
Task completion	5/5 completed all seven tasks; every participant needed assistance on at least one task.	Completion is not the same as independent use.
Independent task performance	Body exploration, stage entry, habitat continuation, media navigation, exit: 5/5 each. Habitat travel: 3/5; food search: 2/5.	T3 and T6 were the main assistance points.
Food-search errors	3/5 participants caught at least one ant; P03 marked 3+ wrong ant catches. P01 had 3+ unsuccessful grab attempts.	Counts recorded as ranges cannot support exact totals.
Knowledge questions	Before: 1/10 correct. After: 9/10 unambiguously correct; P04 marked both correct and incorrect answers to K2.	The short repeated-choice check indicates immediate recognition, not longer-term retention.
Comfort	4/5 no discomfort; 1/5 mild discomfort; none marked moderate/severe.	Short exposure; small sample.

Participant ratings (mean, five-point scale)

Positive indicators	Mean	Friction indicators	Mean
Ladybird control	4.8	XR insect grabbing ease	3.0
Learning cards and video	4.8	Finding habitat without confusion	4.2
Knowing what to do next	4.6	Catch feedback clarity	4.2
Voice guidance and UI	4.6		

Data source: Prototype 2b Test Plan and P01–P05 participant sheets. Original sheets provided separately as appendices.

INTERPRETATION / PATTERNS AND CAUSES

What the evidence means

The most important tensions emerge when observed behaviour and questionnaire ratings are read together.

3 Four evidence-based insights

Insight 1 / Recognition improved, but causal learning remains unproven. Four participants began with 0/2 correct and P02 began with 1/2. Afterwards, all five selected the correct food source, and four gave an unambiguous correct description of aphid shape. Media usefulness also averaged 4.8/5. This supports immediate recognition in the designed setting, not durable biological understanding: the same multiple-choice alternatives were repeated, and cards, narration, video and task exposure were combined. P04's double-marked post-answer further reduces certainty.

Insight 2 / Interaction access, not just insect recognition, was the central failure point. All five eventually reached 10 aphids, but only P02 and P05 did so independently. Grabbing ease was the lowest-rated measure (3.0/5). P01 made 3+ unsuccessful attempts yet recorded no ant confusion, saying the trigger seemed to do nothing; P04 similarly reported that aphids were "hard to point at." In contrast, P03 recorded 3+ incorrect ant catches. These cases point to two different problems: acquiring a reachable or selectable target and correctly identifying it. A larger catch radius alone would not solve conceptual mistakes; extra teaching alone would not repair targeting.

Insight 3 / Confidence in control masked fragile transitions. Ladybird movement rated 4.8/5, yet only 3/5 independently reached the habitat. P02's task row marks T3 as assisted while the observation locates hesitation around the continuation control; P05 also described a pause before the arrival card appeared. This discrepancy suggests the observation protocol did not always isolate navigation from stage handoff. A participant can understand steering and still be uncertain whether the system has recognised arrival. Confirmation must be immediate and unmistakable.

Insight 4 / Player agency and familiarity change what "easy" means. P03 wanted to approach aphids without physically walking; P04 requested controller-based viewpoint movement; P01 wanted the view to follow the ladybird. Conversely, P05 completed catching independently with no unsuccessful grabs and attributed skill to frequent gaming. These comments suggest that viewpoint access and control experience may shape performance. They do not establish causation, especially because every participant reported some previous VR experience.

Design principle emerging from the results

INTERPRETATION A successful XR learning loop needs both understandable content and reliably accessible action. The next task should be visually obvious, physically reachable and confirmed by feedback; otherwise educational progress can be hidden by interaction friction.

JUDGEMENT / DESIGN RESPONSE

What is validated?

Aims are evaluated separately; positive learning scores do not automatically validate every interaction.

4 Evaluation against the original testing intentions

Aim	Judgement	Evidence-based justification
O1 · Learning flow	Mostly supported	Five of five independently completed the core transitions, but P02 reported uncertainty around the habitat prompt and P05 noted delayed arrival feedback.
O2 · Ladybird navigation	Partially supported	Movement was rated 4.8/5, while only 3/5 completed habitat travel without assistance; local handoff issues remain.
O3 · Recognising aphids	Supported for immediate recognition	Unambiguous correct answers increased from 1/10 to 9/10, but repeated questions and combined media limit claims of retained learning.
O4 · Grabbing the right insects	Not yet supported for independent usability	Only 2/5 independently finished catching; grab ease was 3.0/5 and aiming/reach issues recurred.
O5 · Comfortable, clear XR	Partially supported	Four reported no discomfort and one mild discomfort; voice/UI rated 4.6/5, yet control and transition frustration persisted.

5 Updated concept description

The design direction developed from evidence across iterations rather than simply adding features. In IP1, only 1/5 clearly recalled each of the antenna and elytra functions despite positive card-readability ratings. After IP2a introduced multimodal anatomical feedback, its study recorded 15/15 correct immediate anatomy answers; these results cannot be compared causally because the questions and testing contexts differed. IP2a participants also favoured feeding (5/5) and guided movement (4/5), motivating Prototype 2b's habitat and food-search sequence. Tiny Worlds is now a guided, insect-scale XR learning experience: users inspect anatomy, steer a ladybird to its habitat, learn to recognise aphids and choose prey through controller grabbing. The refined concept makes affordances explicit through reachable targets, clear input-mode changes and immediate stage feedback. The current "feeding" task is symbolic collection, not yet a realistic depiction of predation.

6 Targeted design decisions and how to validate them

P1 · Make targets reachable. P01/P03/P04 requested better target access: test bounded locomotion or nearby teleport points. Measure unassisted reach and comfort.

P2 · Make Grip unmistakable. Three needed catch assistance: test clear hover or Grip confirmation and forgiving targeting without automated scoring. Track independent catches and wrong-target attempts.

P3 · Confirm arrival immediately. P02/P05 reported uncertain handoff: display an immediate arrival acknowledgement and persistent, readable Next action. Count stage prompts.

What I learned across the process

Prototype session review

Prototype 2b taught me that completing a task does not necessarily mean the interaction is intuitive. All five participants finished the experience, but only two independently completed the aphid-catching task. Although ladybird movement received 4.8/5, grabbing ease scored only 3.0/5. More importantly, I observed that participants often recognised the aphids but struggled to reach or select them with the controllers. Some repeatedly pointed at the correct targets without a successful response, which caused frustration. Having practised the grabbing interaction myself, I initially assumed it was straightforward. Seeing other users struggle challenged this assumption and made me realise that a design which feels easy to its developer may not be accessible to everyone.

Methodological reflection

Across IP1, IP2a and IP2b, I gradually understood why evaluation is essential to iterative design. In IP1, testing revealed that users often overlooked text-based learning cards, something I had not anticipated. This led me to introduce event-triggered narration and multimodal feedback in IP2a, where immediate knowledge results improved. Without user testing, I would not have identified the original problem or explored a different approach.

In IP2b, combining observation, questionnaires and interviews was especially valuable. Ratings showed where difficulties occurred, while observations revealed that unsuccessful catches often resulted from controller interaction rather than misunderstanding the insects. However, the small sample, repeated knowledge questions and participants' existing VR experience limit the conclusions. Future testing should separately record recognition errors and grabbing failures, include users with different experience levels, and examine whether learning is retained over time.

Concept evaluation

My original XR concept aimed to help users understand insects through immersive spatial exploration rather than passive reading. IP1 revealed limitations in text-based explanations, while IP2a suggested that combining narration, animation and visual feedback could better support immediate understanding. Prototype 2b extended this concept into habitat exploration and food identification.

The current results provide encouraging evidence for the educational approach: knowledge responses improved after the learning activities, and participants generally recognised the aphids. However, recognising a target did not always translate into successfully interacting with it. The main limitation was therefore not insect identification but the accessibility of the VR controls and viewing position. I consider the concept partially validated: its educational potential is promising, but the interaction design needs improvement before the experience can consistently support independent learning.

Improvements and XR extensions

My priority would be to make the experience more accessible rather than simply adding features. Clearer Grip instructions, larger interaction areas, more forgiving selection and controlled movement or teleportation could help users reach aphids without frustration. I would also retain multimodal learning guidance while testing its educational contribution more carefully.

An AR extension could place virtual insects on real plants, connecting learning with authentic habitats. However, small target sizes and environmental tracking might introduce new interaction difficulties. This experience has changed how I approach XR design: I now recognise that successful interaction must be evaluated from different users' perspectives, not only through my own experience as the developer.

Appendix A

Test Plan

Tiny Worlds · Interactive Prototype 2b

PURPOSE Evaluate how well participants understand, navigate and complete the VR learning journey, not how fast they finish.

1 Evaluation aims and evidence

Objective	What this tests	Evidence to collect
O1 · Understand the learning flow	Can players move through body interaction, exploration and learning cards?	Task status; help needed; observed confusion
O2 · Control the ladybird	Can players steer safely to HabitatTarget?	Independent completion; collisions; recovery notes
O3 · Learn about aphids	Does the experience support recognition of aphids and their role as food?	Same two multiple-choice answers before / after; participant explanation
O4 · Identify and grab insects	Can players grip aphids and distinguish them from ants?	10/15 unique aphid target; wrong ant grabs; failed grabs
O5 · Experience VR comfortably	Are UI, narration, video and controller use clear and comfortable?	1–5 ratings; comfort response; participant quotes

The course brief does not require timing. Define success through observable tasks, learning evidence and user feedback. Report both positive and negative cases rather than treating success as guaranteed.

2 Participants and setup

Item	Researcher action
Participants	Aim for 4–6 volunteers; use P01, P02, etc. and do not record names.
Equipment	Use Meta Quest 2 with both controllers and the latest Prototype2b Android build; disable Editor testing bypass.
Method	Moderated observation + multiple-choice BEFORE/AFTER questions + simple ratings + quick interview.
Consistency	Use the same task prompts. Do not disclose which insect is correct before the BEFORE questions.
Safety	Clear the play area, fit the headset, offer a pause/stop at any time, and record discomfort.

3 Test session sequence

BEFORE: Explain the process, obtain consent, check prior VR experience and administer two multiple-choice knowledge questions. Do not reveal the answers.

DURING: Observe tasks T1–T7. Tick one status for each task: Independent, With help, Not completed or Not attempted. Mark issues and write one specific behaviour or participant quote.

AFTER: Repeat the two knowledge questions; let the participant tick 1–5 ratings and comfort. Ask the three Quick Interview questions, recording their own words.

Test procedure

Standard task prompts and consistent observation rules

4 Tasks to give each participant

Step	Neutral moderator instruction	Watch for
T1 · Body exploration	“Choose the ladybird and explore its body parts.”	Understands antenna, wing and elytra interactions.
T2 · Switch modes	“Continue when you are ready.”	Notices and uses Explore Environment without prompting.
T3 · Find habitat	“Guide the ladybird to its habitat.”	Thumbstick control, obstacles, edge recovery and help.
T4 · Continue learning	“Continue to the food learning activity.”	Locates Habitat Continue button and advances.
T5 · Cards and video	“Use the learning cards and video, then proceed.”	Next, Back, audio-video readability; recognises new task.
T6 · Food search	“Complete the food search using the VR controllers.”	Grabbing, 10 unique aphids, mistaken ant grabs, feedback.
T7 · Exit / replay	“Finish the activity and go back to insect selection.”	Finish, Back to Insects, UI clarity and state restoration.

5 Simple scoring and observational notes

For each task, tick only one: I = independent, H = completed with help, N = not completed, — = not attempted. Help includes verbal instructions or physical intervention; do not count the prototype’s built-in guidance as moderator help.

For common issues, tick the matching category (navigation stuck, missed UI, grip difficulty, confusion, sound/video, none) and record a concrete example. For incorrect ant grabs and missed grabs, tick a range; add an exact count only when easy to observe.

For the knowledge check, compare each participant’s BEFORE and AFTER answers. Use the same two questions; note that seeing the BEFORE options may prime users, so avoid overstating causal learning gains.

6 Analysis-ready evidence for Evaluation 3

Evidence	Possible evaluation use
Task status + observed difficulties	Describe what worked, what did not, and which transitions were confusing.
Before/after correct answers	Assess whether aphid learning aims are supported or remain uncertain.
Wrong insect / missed grab ranges	Interpret identification challenge separately from XR handling difficulties.
Ratings + Quick Interview quotes	Explain causes behind positive or negative user experiences.
Method notes / discomfort	Critically evaluate whether the chosen testing method produced useful evidence.

EVALUATION 3 Separate factual results from interpretation. Link conclusions to observations and quotes; reflect on methodology, conceptual validation, improvements and possible AR/XR extensions.

Participant Test Sheet

Prototype 2b · Quick-choice format · No timers or stopwatch needed

Participant: P__01____	**Date / version:** _____23/10_____
VR experience: ☐ None ☒ Some ☐ Frequent	**Participation consent:** ☒ Yes ☐ No

A BEFORE the experience · Tick one answer per question

K1. Which insects do many ladybirds commonly eat?

☐ Butterflies ☐ Aphids ☐ Ants ☒ Not sure

K2. Which feature helps you recognise an aphid?

☐ Narrow waist ☐ Small pear-shaped body ☒ Large colourful wings ☐ Not sure

B During the experience · Tick ONE task result per row

I = Independent H = With help N = Not completed — = Not attempted

Task	Tick one
T1 Select Ladybird + explore body parts	☒ I ☐ H ☐ N ☐ —
T2 Enter environment exploration	☒ I ☐ H ☐ N ☐ —
T3 Move Ladybird to Habitat	☒ I ☐ H ☐ N ☐ —
T4 Continue from Habitat	☒ I ☐ H ☐ N ☐ —
T5 Learning cards + video → food search	☒ I ☐ H ☐ N ☐ —
T6 Catch ANY 10 of 15 aphids	☐ I ☒ H ☐ N ☐ —
T7 Finish → Back to Insects	☒ I ☐ H ☐ N ☐ —

C Quick observation · Tick what you noticed

☐ No issue ☐ Navigation blocked ☐ Missed UI button ☐ Grip difficulty

☐ Unsure what next ☐ Audio/video unclear ☒ Other: __she dosen't know how to grab an aphid._____

Wrong ant grabs: ☒ 0 ☐ 1–2 ☐ 3+ exact (optional): ____

Unsuccessful grab attempts: ☐ 0 ☐ 1–2 ☒ 3+ exact (optional): ____

D Observation notes · What actually happened?

Where / what did the user struggle with, or do well?

_____She didn't know how to grab an aphid; she felt like something was stopping her in front of the aphids, maybe the plant leaf._____

_____

Exact participant words (quote), if possible:

"I don't know why this is not working. I feel like I point at that aphid, but when I press the trigger, nothing happens."_____

_____

After the experience

Participant ID: P__01__ · Remove headset before answering

E AFTER knowledge check · Tick one answer per question

K1. Which insects do many ladybirds commonly eat?

☐ Butterflies ☒ Aphids ☐ Ants ☐ Not sure

K2. Which feature helps you recognise an aphid?

☐ Narrow waist ☒ Small pear-shaped body ☐ Large colourful wings ☐ Not sure

Researcher only: BEFORE correct: _0_ / 2 AFTER correct: __2_ / 2

F Short participant ratings · Tick one box per statement

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	Choose one
Q1 I knew what to do next at each stage.	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5
Q2 Moving the ladybird was easy to control.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q3 I could find the habitat without confusion.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q4 The cards and video helped me recognise aphids.	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5
Q5 Catching insects with the Quest controllers was easy.	☐ 1 ☒ 2 ☐ 3 ☐ 4 ☐ 5
Q6 I understood why a catch was correct or incorrect.	☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5
Q7 The voice guidance and UI were clear.	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5

Comfort in VR: ☒ No discomfort ☐ Mild ☐ Moderate ☐ Severe

G Quick Interview · Ask aloud and write a short answer

1. At what point, if any, were you unsure what to do next?

_move to explore_____

_____

2. What helped you recognise aphids? What was confusing?

_ _ Colors are different in the environment; I don't know how to catch them.

_____

_____

3. If you could change one thing in Tiny Worlds, what would it be?

_to let player view follow the ladybird_____

_____

H Researcher conclusion (one evidence-based note)

Most important finding: _____the participant has difficulty catching aphids in the section._____

Issue severity: ☐ None ☐ Low ☒ Medium ☐ High

Supporting evidence (task, response or quote):

"I don't know why I press the trigger it's not working."_____

Participant Test Sheet

Prototype 2b · Quick-choice format · No timers or stopwatch needed

Participant: P__02____	**Date / version:** 23 October 2026
VR experience: ☐ None ☒ Some ☐ Frequent	**Participation consent:** ☒ Yes ☐ No

A BEFORE the experience · Tick one answer per question

K1. Which insects do many ladybirds commonly eat?

☐ Butterflies ☒ Aphids ☐ Ants ☐ Not sure

K2. Which feature helps you recognise an aphid?

☐ Narrow waist ☐ Small pear-shaped body ☐ Large colourful wings ☒ Not sure

B During the experience · Tick ONE task result per row

I = Independent H = With help N = Not completed — = Not attempted

Task	Tick one
T1 Select Ladybird + explore body parts	☒ I ☐ H ☐ N ☐ —
T2 Enter environment exploration	☒ I ☐ H ☐ N ☐ —
T3 Move Ladybird to Habitat	☐ I ☒ H ☐ N ☐ —
T4 Continue from Habitat	☒ I ☐ H ☐ N ☐ —
T5 Learning cards + video → food search	☒ I ☐ H ☐ N ☐ —
T6 Catch ANY 10 of 15 aphids	☒ I ☐ H ☐ N ☐ —
T7 Finish → Back to Insects	☒ I ☐ H ☐ N ☐ —

C Quick observation · Tick what you noticed

☐ No issue ☐ Navigation blocked ☒ Missed UI button ☐ Grip difficulty

☒ Unsure what next ☐ Audio/video unclear ☐ Other: _____

Wrong ant grabs: ☐ 0 ☒ 1–2 ☐ 3+ exact (optional): 1

Unsuccessful grab attempts: ☐ 0 ☒ 1–2 ☐ 3+ exact (optional): 2

D Observation notes · What actually happened?

Where / what did the user struggle with, or do well?

The participant completed most tasks independently. They briefly missed the continue/navigation UI after reaching the habitat and needed help once before continuing.

Exact participant words (quote), if possible:

"I can see where I am going, but I am not sure what I should press next."

After the experience

Participant ID: P 02 · Remove headset before answering

E AFTER knowledge check · Tick one answer per question

K1. Which insects do many ladybirds commonly eat?

☐ Butterflies ☒ Aphids ☐ Ants ☐ Not sure

K2. Which feature helps you recognise an aphid?

☐ Narrow waist ☒ Small pear-shaped body ☐ Large colourful wings ☐ Not sure

Researcher only: BEFORE correct: 1 / 2 AFTER correct: 2 / 2

F Short participant ratings · Tick one box per statement

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	Choose one
Q1 I knew what to do next at each stage.	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5
Q2 Moving the ladybird was easy to control.	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5
Q3 I could find the habitat without confusion.	☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5
Q4 The cards and video helped me recognise aphids.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q5 Catching insects with the Quest controllers was easy.	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5
Q6 I understood why a catch was correct or incorrect.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q7 The voice guidance and UI were clear.	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5

Comfort in VR: ☐ No discomfort ☒ Mild ☐ Moderate ☐ Severe

G Quick Interview · Ask aloud and write a short answer

1. At what point, if any, were you unsure what to do next?

After moving the ladybird to the habitat, I was briefly unsure which UI control would continue the experience.

2. What helped you recognise aphids? What was confusing?

The learning cards and video helped, especially the small pear-shaped body. From a distance I briefly confused an ant with an aphid.

3. If you could change one thing in Tiny Worlds, what would it be?

I would make the next-step button or prompt more noticeable when moving between stages.

H Researcher conclusion (one evidence-based note)

Most important finding: Navigation was generally clear, but the habitat transition needed a stronger next-step cue.

Issue severity: ☐ None ☒ Low ☐ Medium ☐ High

Supporting evidence (task, response or quote):

T3 required help once; the participant also reported being unsure which control to press after reaching the habitat.

Participant Test Sheet

Prototype 2b · Quick-choice format · No timers or stopwatch needed

Participant: P__03____	**Date / version:** _____28 October 2026_____
VR experience: ☐ None ☒ Some ☐ Frequent	**Participation consent:** ☒ Yes ☐ No

A BEFORE the experience · Tick one answer per question

K1. Which insects do many ladybirds commonly eat?

☐ Butterflies ☐ Aphids ☐ Ants ☒ Not sure

K2. Which feature helps you recognise an aphid?

☐ Narrow waist ☐ Small pear-shaped body ☐ Large colourful wings ☒ Not sure

B During the experience · Tick ONE task result per row

I = Independent H = With help N = Not completed — = Not attempted

Task	Tick one
T1 Select Ladybird + explore body parts	☒ I ☐ H ☐ N ☐ —
T2 Enter environment exploration	☒ I ☐ H ☐ N ☐ —
T3 Move Ladybird to Habitat	☒ I ☐ H ☐ N ☐ —
T4 Continue from Habitat	☒ I ☐ H ☐ N ☐ —
T5 Learning cards + video → food search	☒ I ☐ H ☐ N ☐ —
T6 Catch ANY 10 of 15 aphids	☐ I ☒ H ☐ N ☐ —
T7 Finish → Back to Insects	☒ I ☐ H ☐ N ☐ —

C Quick observation · Tick what you noticed

☐ No issue ☐ Navigation blocked ☐ Missed UI button ☒ Grip difficulty

☐ Unsure what next ☐ Audio/video unclear ☐ Other: _____

Wrong ant grabs: ☐ 0 ☐ 1–2 ☒ 3+ exact (optional): ____

Unsuccessful grab attempts: ☐ 0 ☒ 1–2 ☐ 3+ exact (optional): ____

D Observation notes · What actually happened?

Where / what did the user struggle with, or do well?

The participant want to get close to the aphids without physically moving around.

_____

_____

_____

Exact participant words (quote), if possible:

"Do I need to walk to get close to those aphids?"_____

_____

After the experience

Participant ID: P___03___ · Remove headset before answering

E AFTER knowledge check · Tick one answer per question

K1. Which insects do many ladybirds commonly eat?

☐ Butterflies ☒ Aphids ☐ Ants ☐ Not sure

K2. Which feature helps you recognise an aphid?

☐ Narrow waist ☒ Small pear-shaped body ☐ Large colourful wings ☐ Not sure

Researcher only: BEFORE correct: __0_ / 2 AFTER correct: _2_ / 2

F Short participant ratings · Tick one box per statement

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	Choose one
Q1 I knew what to do next at each stage.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q2 Moving the ladybird was easy to control.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q3 I could find the habitat without confusion.	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5
Q4 The cards and video helped me recognise aphids.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q5 Catching insects with the Quest controllers was easy.	☐ 1 ☒ 2 ☐ 3 ☐ 4 ☐ 5
Q6 I understood why a catch was correct or incorrect.	☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5
Q7 The voice guidance and UI were clear.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5

Comfort in VR: ☒ No discomfort ☐ Mild ☐ Moderate ☐ Severe

G Quick Interview · Ask aloud and write a short answer

1. At what point, if any, were you unsure what to do next?

When kept grabbing ants_____
_____

2. What helped you recognise aphids? What was confusing?

They are vibrant.__nothing feel confused this part_____

3. If you could change one thing in Tiny Worlds, what would it be?

_Have skip or walk around so can reach aphids_____
_____

H Researcher conclusion (one evidence-based note)

Most important finding: ____should make the aphids easier to grab to reduce the participant's frustration._____

Issue severity: ☐ None ☐ Low ☒ Medium ☐ High

Supporting evidence (task, response or quote):

_it feels hard to pointing at an aphids._____

Participant Test Sheet

Prototype 2b · Quick-choice format · No timers or stopwatch needed

Participant: P__04____	**Date / version:** ___27 October 2026_____
VR experience: ☐ None ☒ Some ☐ Frequent	**Participation consent:** ☒ Yes ☐ No

A BEFORE the experience · Tick one answer per question

K1. Which insects do many ladybirds commonly eat?

☐ Butterflies ☐ Aphids ☐ Ants ☒ Not sure

K2. Which feature helps you recognise an aphid?

☐ Narrow waist ☐ Small pear-shaped body ☐ Large colourful wings ☒ Not sure

B During the experience · Tick ONE task result per row

I = Independent H = With help N = Not completed — = Not attempted

Task	Tick one
T1 Select Ladybird + explore body parts	☒ I ☐ H ☐ N ☐ —
T2 Enter environment exploration	☒ I ☐ H ☐ N ☐ —
T3 Move Ladybird to Habitat	☒ I ☐ H ☐ N ☐ —
T4 Continue from Habitat	☒ I ☐ H ☐ N ☐ —
T5 Learning cards + video → food search	☒ I ☐ H ☐ N ☐ —
T6 Catch ANY 10 of 15 aphids	☐ I ☒ H ☐ N ☐ —
T7 Finish → Back to Insects	☒ I ☐ H ☐ N ☐ —

C Quick observation · Tick what you noticed

☐ No issue ☐ Navigation blocked ☐ Missed UI button ☐ Grip difficulty

☐ Unsure what next ☐ Audio/video unclear ☐ Other: _hard to grab aphids using controller_____

Wrong ant grabs: ☒ 0 ☐ 1–2 ☐ 3+ exact (optional): ____

Unsuccessful grab attempts: ☐ 0 ☒ 1–2 ☐ 3+ exact (optional): ____

D Observation notes · What actually happened?

Where / what did the user struggle with, or do well?

__she knows how to go through the whole process, but the struggling part is how to grab aphids using controller._____

_____

_____

Exact participant words (quote), if possible:

“The aphids feel hard to point at.”_____

_____

After the experience

Participant ID: P___04__ · Remove headset before answering

E AFTER knowledge check · Tick one answer per question

K1. Which insects do many ladybirds commonly eat?

☐ Butterflies ☒ Aphids ☐ Ants ☐ Not sure

K2. Which feature helps you recognise an aphid?

☐ Narrow waist ☒ Small pear-shaped body ☒ Large colourful wings ☐ Not sure

Researcher only: BEFORE correct: _0_ / 2 AFTER correct: __2_ / 2

F Short participant ratings · Tick one box per statement

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	Choose one
Q1 I knew what to do next at each stage.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q2 Moving the ladybird was easy to control.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q3 I could find the habitat without confusion.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q4 The cards and video helped me recognise aphids.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q5 Catching insects with the Quest controllers was easy.	☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5
Q6 I understood why a catch was correct or incorrect.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q7 The voice guidance and UI were clear.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5

Comfort in VR: ☒ No discomfort ☐ Mild ☐ Moderate ☐ Severe

G Quick Interview · Ask aloud and write a short answer

1. At what point, if any, were you unsure what to do next?

_____not really_____

_____

2. What helped you recognise aphids? What was confusing?

_____the light green color against the nature's green color_____

_____

3. If you could change one thing in Tiny Worlds, what would it be?

_____that I could move using the controllers_____

_____

H Researcher conclusion (one evidence-based note)

Most important finding: ____the participant want to find a way to control the view_____

Issue severity: ☐ None ☐ Low ☒ Medium ☐ High

Supporting evidence (task, response or quote):

It's hard to get close to the aphids._____

Participant Test Sheet

Prototype 2b · Quick-choice format · No timers or stopwatch needed

Participant: P__05____	**Date / version:** ___29/10/2026_____
VR experience: ☐ None ☒ Some ☐ Frequent	**Participation consent:** ☒ Yes ☐ No

A BEFORE the experience · Tick one answer per question

K1. Which insects do many ladybirds commonly eat?

☐ Butterflies ☐ Aphids ☐ Ants ☒ Not sure

K2. Which feature helps you recognise an aphid?

☐ Narrow waist ☐ Small pear-shaped body ☒ Large colourful wings ☐ Not sure

B During the experience · Tick ONE task result per row

I = Independent H = With help N = Not completed — = Not attempted

Task	Tick one
T1 Select Ladybird + explore body parts	☒ I ☐ H ☐ N ☐ —
T2 Enter environment exploration	☒ I ☐ H ☐ N ☐ —
T3 Move Ladybird to Habitat	☐ I ☒ H ☐ N ☐ —
T4 Continue from Habitat	☒ I ☐ H ☐ N ☐ —
T5 Learning cards + video → food search	☒ I ☐ H ☐ N ☐ —
T6 Catch ANY 10 of 15 aphids	☒ I ☐ H ☐ N ☐ —
T7 Finish → Back to Insects	☒ I ☐ H ☐ N ☐ —

C Quick observation · Tick what you noticed

☒ No issue ☐ Navigation blocked ☐ Missed UI button ☐ Grip difficulty

☐ Unsure what next ☐ Audio/video unclear ☐ Other: _____

Wrong ant grabs: ☐ 0 ☒ 1–2 ☐ 3+ exact (optional): ____

Unsuccessful grab attempts: ☒ 0 ☐ 1–2 ☐ 3+ exact (optional): ____

D Observation notes · What actually happened?

Where / what did the user struggle with, or do well?

_____The user is quite good at the catch aphids part, I'm impressed.

Exact participant words (quote), if possible:

_____interviewer: _" Why are you so good at catching?"_____interviewee_: "I play games frequently."

After the experience

Participant ID: P___05___ · Remove headset before answering

E AFTER knowledge check · Tick one answer per question

K1. Which insects do many ladybirds commonly eat?

☐ Butterflies ☒ Aphids ☐ Ants ☐ Not sure

K2. Which feature helps you recognise an aphid?

☐ Narrow waist ☒ Small pear-shaped body ☐ Large colourful wings ☐ Not sure

Researcher only: BEFORE correct: _0_ / 2 AFTER correct: _2_ / 2

F Short participant ratings · Tick one box per statement

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	Choose one
Q1 I knew what to do next at each stage.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q2 Moving the ladybird was easy to control.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q3 I could find the habitat without confusion.	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5
Q4 The cards and video helped me recognise aphids.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q5 Catching insects with the Quest controllers was easy.	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5
Q6 I understood why a catch was correct or incorrect.	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5
Q7 The voice guidance and UI were clear.	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5

Comfort in VR: ☒ No discomfort ☐ Mild ☐ Moderate ☐ Severe

G Quick Interview · Ask aloud and write a short answer

1. At what point, if any, were you unsure what to do next?

____maybe better use some guide animation to make it more clear to use._____

2. What helped you recognize aphids? What was confusing?

_____no confuse, the shape and color of the aphids is obvious

_____

_____

3. If you could change one thing in Tiny Worlds, what would it be?

_____maybe make two kinds of insets fight tother can improve the interests, Or it could be played online against opponents, which would make it more suitable for commercialization._____

H Researcher conclusion (one evidence-based note)

Most important finding: ___when the user reach the target habitat, it won't triger the next event card immediately. It has a time delay.

_____

Issue severity: ☐ None ☐ Low ☒ Medium ☐ High

Supporting evidence (task, response or quote):

__Why is there no change when I reach the position?___"_____